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Reference-Free Human-Automation Shared Control for Obstacle Avoidance of Automated Vehicles

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***Abstract*—In this paper, a novel reference-free shared control system is designed for obstacle avoidance for automated vehicles. Rather than using a reference path to guide the driver, the proposed framework constrains the vehicle's status to guarantee the safety without scarifying the driver's freedom. The constrained Delaunay triangle method is introduced to identify the vehicle's position constraints and the constraints of obstacle avoidance, vehicle stability and physical limitations are investigated and unified. A nonlinear predictive control problem, which is constructed accounting nonlinear vehicle dynamics and given driver actions, is designed to optimize the steering and braking actions needed to keep the vehicle safe. The automation is supposed to correct the driver's steering or braking actions to prevent constraint violation and losing the control of vehicle. The simulation results show that the automation can assist the driver to avoid obstacles and guarantee the vehicle's stability with minimal control intervention.**

***Index Terms*—Human-automation collaboration, shared control, automated vehicle**

I. Introduction

Automated vehicles have the potential to increase road capacity and reduce fuel consumption. Even though the autonomous driving can decrease the chance of vehicle accidents caused by human driver (drowsiness, gaze), there are still many factors preventing the widespread of fully automated driving vehicles [1], [2]. Many studies report that high automation can decrease vigilance and increase boredom. The reliance on automation may render human driver as unnecessary [3]. The automated vehicle may not be fully adopted by the majority of the population in the foreseeable future until further testing could be completed.

To bridge the gaps mentioned above, the shared control approach is proposed. The idea is to make human driver work with automation (planning, decision making, action execution) to achieve a common goal [4], [5], [6]. Based on the types of Human Machine Interface, the shared control system can be divided into visual, auditory shared control, haptic and multi-modal interface shared control system. Compared with visual and auditory shared control [7], haptic shared control is a promising approach as it enables the human driver to communicate continuously with automation through a haptic interface [8], [9]. Haptic shared control can lead to performance benefits such as faster response and less control activity needed [10], [11]. The multi-modal interface shared control has drawn much attention as it is expected to provide a more natural and less distractive interaction [12].

Based on the driving task, the shared control system can be divided into path-based approaches and reference-free approaches. The main difference between them is that there is a given reference trajectory in the path-based approaches. This reference trajectory can be generated by some common path planning methods such as model predictive control (MPC) [13], rapidly exploring random trees (RRTs) [14] and geometric method [15]. The automation is supposed to guide the driver to follow this reference trajectory [16]. This approach has been widely studied because of its controllability and feasibility [8]. However, some critical issues are needed to be further explored such as the conflict between the driver and the automation. Who has the main authority to execute the driving tasks is the most significant question in the decision-making process [17]. The solutions of this problem are based on the mutual understanding between the human driver, automation and traffic conditions [18]. In addition, some scholars question the superiority and rationality of the reference trajectory [19]. A non-adaptive reference trajectory may not meet the driver's expectations. In view of all this, the reference-free approach is investigated under the premise of no given reference path nor trajectory. The idea is to make the automation work as a supervisor. The automation only intervenes when necessary such as near-crash situations and loss of stability [20]. In this way, the human driver retains as much direct control freedom as possible and feels the assisted torque generated by the

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automation in an emergency situation [21].

In this paper, we propose a reference-free based shared control system for collision avoidance for automated driving vehicle. Collision avoidance is one of the most important manoeuvres for autonomous driving, and the causes of collisions are related to driver condition or improper action such as driver inexperience, inattention and passing improperly at most time [22]. The collision can be divided into head-on collision, rear-collision and side collision. In this paper, only head-on collision (longitudinal) and side collision (lateral) are considered. Current technology such as Automatic braking system (ABS) and pedestrian monitors are developed and employed to warn the driver against potential collisions [23]. Besides the visual, auditory and physical cues, we focus on the design of a haptic interface through which the driver can feel the assistance by the automation and then execute an appropriate control action. Without knowing the human driver's objective, we utilize the safety constraints to guarantee the safety and stability of the vehicle system in the driving task. In particular, constrained Delaunay triangles are formulated to identify the lateral safe area, and time to collision (TTC) is utilized for longitudinal collisions avoidance. In addition, the stability constraints such as sideslip angle and yaw rate and vehicle handling limits are considered to eliminate the occurrence of oversteer and understeer. Furthermore, a predictive control optimization problem is formulated, which simultaneously uses the predicted driver behavior and determines the minimum intervention torque that keeps the vehicle in a safe region. The nonlinear vehicle dynamics is considered in the predictive control optimization problem without scarifying the complexity of vehicle dynamics.

The remainder of the paper is structured as follows. In Section II, the vehicle model is given. The main contributions of this paper consists of identification of safety constraints are introduced in Section III. The prediction optimization problem is formulated in Section IV. The simulation results are demonstrated in Section V. Section VI concludes this paper.

II. System Modeling

The dynamics of the vehicle is modeled by the following set of differential equations (Fig 1) [24]:

$$\begin{aligned} \dot{v}_y &= -r v_x + \frac{2}{m}(F_{cf}\cos\delta_f + F_{cr}), \\ \dot{r} &= \frac{2}{I_z}(l_f F_{cf} - l_r F_{cr}), \\ \dot{X} &= v_x\cos\psi - v_y\sin\psi, \\ \dot{Y} &= v_x\sin\psi + v_y\cos\psi. \end{aligned} \tag{1}$$

where x and y are the coordinates of the centre of mass (CM) in an inertial frame. v_x is the longitudinal speed and assumed as a constant. $v_y = \dot{y}$ is the lateral speeds of CM in an inertial frame. ψ is the inertial heading angle and $r = \dot{\psi}$ is the yaw rate. m and I_z denote the vehicle's mass and yaw inertia, respectively. l_f and l_r represent the distance from CM of the vehicle to the front and rear axles, respectively. $\dot{X}$ and $\dot{Y}$ denote the longitudinal and lateral speeds of CM in the body frame (X, Y), respectively.

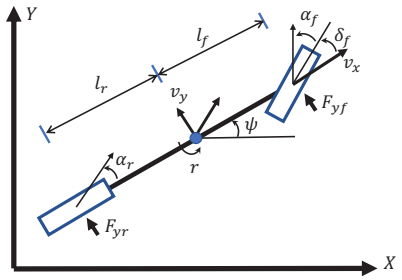


Fig. 1: The vehicle dynamics model

F_{cf} and F_{cr} denote the lateral tyre forces at the front and rear wheels, respectively. $F_{ci}, (i = f, r)$ can be calculated using a linear tire model:

$$\begin{aligned} F_{cf} &= -C_{\alpha f}\alpha_f \\ F_{cr} &= -C_{\alpha r}\alpha_r \end{aligned} \tag{2}$$

where $C_{\alpha f}$ and $C_{\alpha r}$ are the stiffness coefficients for front tire and rear tire. $\alpha_i, (i = f, r)$ denotes tire slip angle, which can be expressed as:

$$\begin{aligned} \alpha_f &\approx \frac{v_y + l_f r}{v_x} - \delta_f \\ \alpha_r &\approx \frac{v_y - l_r r}{v_x}. \end{aligned} \tag{3}$$

Combing (1), (2) and (3), we can have the following compact form of vehicle dynamics:

$$\dot{x}(t) = f(x(t), u(t)). \tag{4}$$

where $x = [X,\ Y,\ r,\ v_y,\ \psi]^T$ is the system state vector. The front wheel angle consists of two parts: $\delta_f = \delta_d + \delta_a$, in which, δ_d is the given driver-commanded steering angle while δ_a is the corrected steering angle provided by the automation. Therefore, systems' input signal is the front wheel angle δ_a. The generation of corrective control actions will be introduced in Section IV.

III. Safety constraints

In this section, we describe the limitations on the execution of driving task in details.

A. Collision avoidance

1) Lateral collision avoidance: In the driving region, the information of traffic is assumed known by the current automotive sensor technology. The driving region can be discretized into constrained Delaunay triangles. Each sequence of adjacent Delaunay triangles which connect the start position S_p with a goal point/goal region F_p can constitute a channel. Each channel consists of many possible collision-free paths (Fig. 2 for example).

Let $T_i, i = 1, \cdots, N$ denote a constructed triangle in the driving region. We note that N varies with time as the driving

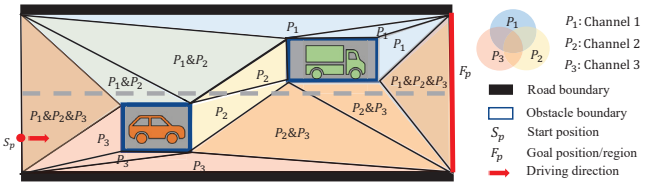


Fig. 2: Driving region with constraint triangles

environment changes. Let $\mathcal{T}$ be the set of all the triangles in the region, i.e., $\mathcal{T} = \{T_1, T_2, \cdots, T_N\}$. Introduce $t \in \mathcal{T}$ as a variable. Denote P as a channel from S_p to F_p, which is a sequence of triangles, i.e., $P = (t_1, t_2, \cdots, t_n) \in \mathcal{T} \times \mathcal{T} \times \cdots \times \mathcal{T}$. We note that in a driving region, there may exist multiple channels. Thus, the numbers of triangles of different channels, i.e., n, may be different. An optimal channel with a sequence of triangles is obtained by evaluating the goodness of each possible channel which can be formulated as the following optimization problem:

$$\begin{aligned} &\min_{t_1,\cdots,t_n} \sum_{i=1}^{n} \{\alpha L_{i,i+1} + \delta \phi_{i,i+1}\} \\ &s.t. t_i \in \mathcal{T} \end{aligned} \tag{5}$$

where α and δ are the corresponding weighting factors. The above optimization problem (5) can be effectively solved by Dijkstra's algorithm [25]. L_i and ϕ represent the length and smoothness of each channel, respectively, which are described below.

1) **Length**. For a specific chanel (e.g. $P = (t_1, t_2, \cdots, t_n)$), the length is used to describe the average distance traveled by all $t_i \in \mathcal{T}$ and is calculated by

$$L_P = \sum_{i=1}^{n} L_{i,i+1} \tag{6}$$

where $e_{i-1,i}$ is the common edge between triangle t_i and t_{i+1}. $M_{i-1,i}$ is the midpoint on the edge $e_{i-1,i}$ and $L_{i-1,i}$ is the length connecting $M_{i-2,i-1}$ with $M_{i-1,i}$ (see Fig. 3).

2) **Smoothness**. Similar with Turn number, the path generated in the channel P is required to be as smooth as possible by considering the driving comfort. We use the orientation of the vehicle to describe the smoothness of the triangles, i.e.,

$$S_P = \sum_{i=1}^{n} \phi_{i-1,i} \tag{7}$$

where ϕ is the change of heading angle between two adjacent triangle (Fig. 4).

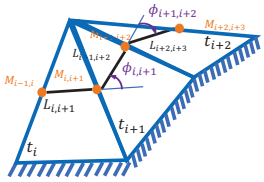


Fig. 4: Smoothness of the Channel P

Once the optimal channel $P_\star$ is identified, the physical positions of its edges can be enforced as lateral position constraints of the ego vehicle which can be compactly written as the following linear inequality:

$$y_{min}(k) + d_b \leqslant y(k) \leqslant y_{max}(k) - d_b \tag{8}$$

where y_{min} and y_{max} represent the lower and upper bound of the ego vehicle's position at each time step k, respectively, and d_b is a minimum safe distance. Condition (8) is compactly written as

$$H(x, \delta_d, u) \leqslant \mathbf{0} \tag{9}$$

2) Longitudinal collision avoidance: In this paper, we introduce the time to collision (TTC) as a decision-making aid to avoid the longitudinal collisions [26]. In a potential head-on collision situation, the calculation of TTC is defined as

$$TTC = \frac{d_r}{v_x} \tag{10}$$

where v_x is the ego vehicle's speed and d_r is the relative distance between the predicted position of ego vehicle and forward obstacle/vehicle in longitudinal direction. It should be mentioned that the TTC should be computed every time step and satisfy

$$TTC \geqslant TTC_{min} \tag{11}$$

The value of TTC_{min} depends on the driving scenario, vehicle's velocity, acceleration etc. The larger the value of

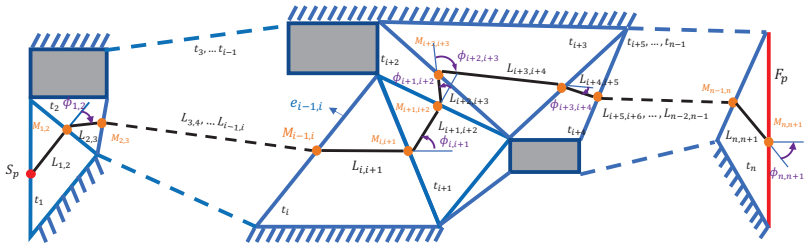


Fig. 3: Length of the Channel P

TTC_{min} is, the more intense actions should be applied and vice versa. Constraint (11) can be rewritten as

$$T(x, \delta_d, u) \leqslant \mathbf{0}. \tag{12}$$

B. Stability constraints

In this paper, the vehicle stability is guaranteed through constraints on the front tire slip angle and yaw rate [16]. The bounds of yaw rate can be identified as

$$||r|| \leqslant r_{max} = \frac{g\mu}{v_x} \tag{13}$$

where g is the gravitational constant and μ is the surface coefficient of friction.

Based on [27], the maximum sideslip β can be represented as

$$||\beta|| \leqslant \beta_{max} = \tan^{-1}(3\frac{mg\mu}{C_r}\frac{l_f}{l_f + l_r}) + \frac{l_r r}{v_x} \tag{14}$$

In summary, the constraints (13) and (14) can be compactly written as

$$S(x, \delta_d, u) \leqslant \mathbf{0}. \tag{15}$$

IV. Predictive control problem formulation

Here, we introduce a reference-free shared control system for autonomous driving. As mentioned before, the input signals of the vehicle are driver-commanded steering angle δ_f and corrected steering angle δ_a. Once the vehicle's predicted states violate the safety constraints introduced in Section III, a corrected steering angle δ_a is generated and injected to the vehicle to guarantee the safety of vehicle.

We discretize system (4) with a fixed sampling time T_p to obtain

$$x_{k+1} = f^d(x_k, u_k, \delta_d) \tag{16}$$

The optimal predictive control problem can be expressed as

$$\min_{\mathcal{U}_t} \sum_{k=0}^{H_c-1} \|u_{t+k,t}\|_Q^2 + \|\Delta u_{t+k,t}\|_R^2 \tag{17a}$$

$$s.t.\ \ x_{t+k+1,t} = f^d(x_{t+k,t}, u_{t+k,t}, \delta_d),\ k = 0, \cdots, H_p - 1 \tag{17b}$$

$$u_{t+k,t} = \Delta u_{t+k,t} + u_{t+k-1,t} \tag{17c}$$

$$u_{min} \leqslant u_{t+k,t} \leqslant u_{max},\ k = 0, \cdots, H_c - 1 \tag{17d}$$

$$\Delta u_{min} \leqslant \Delta u_{t+k,t} \leqslant \Delta u_{max},\ k = 0, \cdots, H_c - 1 \tag{17e}$$

$$H_t(x_{t+k,t}, u_{t+k,t}) \leqslant \mathbf{0}, k = 0, \cdots, H_p, \tag{17f}$$

$$T_t(x_{t+k,t}, u_{t+k,t}) \leqslant \mathbf{0}, k = 0, \cdots, H_p, \tag{17g}$$

$$S_t(x_{t+k,t}, u_{t+k,t}) \leqslant \mathbf{0}, k = 0, \cdots, H_p, \tag{17h}$$

$$\Delta u_{t+k,t} = 0,\ k = H_c, \cdots, H_p \tag{17i}$$

$$u_{t-1,t} = u(t-1) \tag{17j}$$

$$x_{t,t} = x(t) \tag{17k}$$

where t denotes the current time step and $x_{t+k,t}$ denotes the predicted state at time $t + k$ obtained by applying control sequence $\mathcal{U}_t = [u_{t,t}, \cdots, u_{t+k,t}]$ to system (16) with $x_{t,t} = x(t)$. Q, R and ρ are weight factors of control actions, change rate of control and violation of soft constraints, respectively. H_c is the control horizon while H_p is the prediction horizon with $H_p \geqslant H_c$. The control inputs is $u = \begin{bmatrix}\delta_a\end{bmatrix}$.

It should be mentioned that there is no reference path/trajectory imposed in the cost function (17a). The objective of the optimization problem is to minimize control action without violating the safety constraints (17d)-(17h). If the driver is capable of steering the vehicle with driver commands δ_d, no corrective control action will be applied ($\delta_a = 0$) and the cost function will thus be zero. On the contrary, if the driver cannot handle the driving situation and violate any constraints of (17d)-(17h), a corrective control action is generated by the optimization problem (17a) and implemented to the vehicle.

The corrective control actions and the changes of them should satisfy

$$u_{min} = \delta_{amin} \leqslant u_{t+k,t} \leqslant \delta_{amax} = u_{max}$$

and

$$\Delta u_{min} = \Delta\delta_{amin} \leqslant \Delta u_{t+k,t} \leqslant \Delta\delta_{amax} = \Delta u_{max}$$

where δ_{amin}, δ_{amax}, $\Delta\delta_{amin}$ and $\Delta\delta_{amax}$ denote the saturation level, respectively.

The constraint (17b) represents the vehicle dynamics with a given driver's command δ_d. The nonlinear constraints (17f)-(17h) reflect lateral collision avoidance, longitudinal collision avoidance and stability constraints, respectively.

V. Simulation validation

The driving scenario is a two-way freeway. An ego vehicle drives from the start position $(0, 0)$ with a constant speed $(10m/s)$. On the premise of safety, the driver would like to change the lane and overtake the forward vehicle which with a low speed in order to reach to the goal area as soon as possible. The automation here is implemented to guide the driver to achieve the driving task.

A. Simulation environment

The simulation environment is set as follows:

1) The limitation on the control action and change of control action are set as
$$-40 \times \frac{\pi}{180} \leqslant \delta_a \leqslant 40 \times \frac{\pi}{180}$$
$$-20 \times \frac{\pi}{180} \leqslant \Delta\delta_a \leqslant 20 \times \frac{\pi}{180}$$
2) The sample time is $0.1s$. The control horizon is $H_c = 5$ while prediction horizon is $H_p = 10$.
3) The minimum TTC is set as $0.25s$.
4) The road width is $3.5m$. The width of the vehicles change range from $1.5m$ to $2.5m$.
5) Q and R are set as identity matrix with appropriate dimensions.
6) The minimum safe distance $d_b = 1m$.
7) The parameters of vehicle are given by TABLE I.

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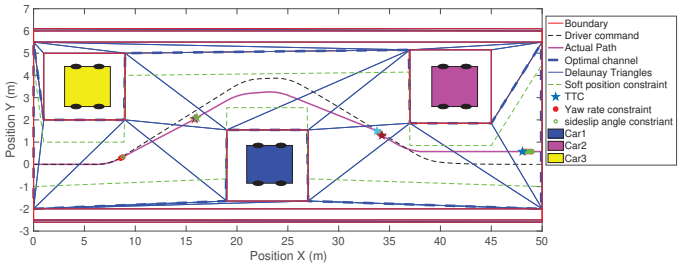
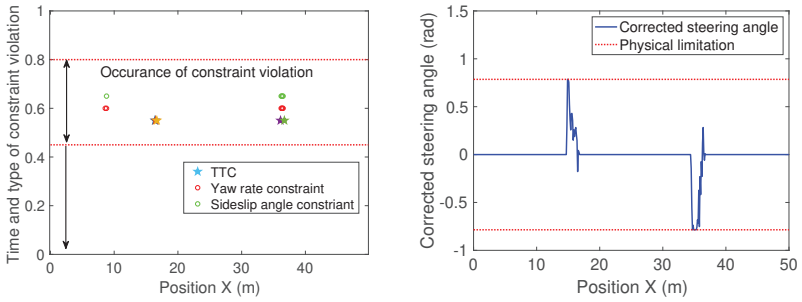
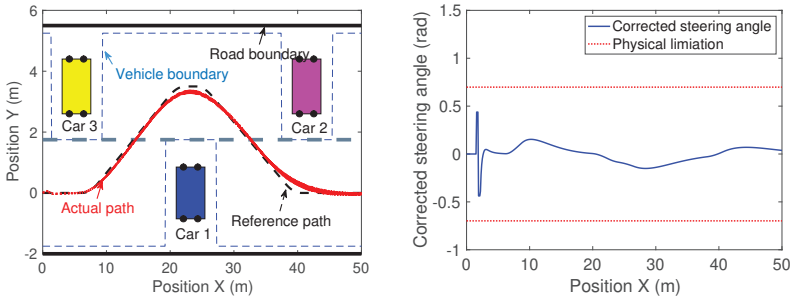


Fig. 5: Simulation result with the proposed approach



(a) (b)

Fig. 6: (a) Occurrence of constraints violation. (b) The correction actions provided by the proposed approach.



(a) (b)

Fig. 7: (a) Tracking performance of the path-based approach. (b) The correction actions provided by path-based approach.

TABLE I: Vehicle Parameters

Parameter	Value
l_f	$1.21m$
l_r	$1.05\ m$
m	$2000\ kg$
I_z	$1300\ kg \cdot m^2$
μ	0.5

B. Simulation results

We compare with a path-based approach to demonstrate the superiority of the proposed approach. In the path-based approach, the reference path is generated by the geometric method and a sequence of driver's steering angle is given. More details can be found in [28].

In Fig. 5, the black dash line represents the driving path based on the given driver's steering angle (driver's path for short) and the magenta solid line represents the actual path with correction actions (steering angle). The Fig. 6a shows the occurrence of the constraint violation. Although the driver's path is a collision-free path, the driver's behavior violates the safety constraints which may induce oversteer/understeer and the position constraints during the turning. Consequently, the predictive controller corrects the driver's control actions (shown in Fig. 6b) to avoid the constraint violation. After the vehicle path has been corrected, the control actions vanish, and the driver's steering angle is again the only input of the vehicle control.

The Fig. 7a shows the vehicle path provided by the baseline approach, i.e. the path-based one, with the same given driver's steering angle, and the Fig. 7b represents the corresponding corrected actions. Compared with the path-based approach

(Fig. 7a), the vehicle motion has been slightly corrected (Fig. 6b) to ensure the stable operating conditions. The path-based approach has to continuously correct the driver's behavior by imposing steering angle and braking torque. It is also noted that due to the prediction of vehicle's states, the automation can predict the potential collision or constraint violation ahead of time. The automation prefers to correct the driver's steering instead of braking.

It should be mentioned that the longer the prediction horizon is, the earlier the automation take actions and the better the driving performance. However, a long prediction horizon brings with more computational burden and increases simulation time. In addition, we notice that with the increase of *minimum TTC*, the automation can detect the head-on obstacle more timely and execute an appropriate control action immediately.

VI. Conclusion

This paper has presented a novel shared control method to assist driver in the execution of driving task with minimal machine intervention. By identifying the constraints on safety and stability and unifying them into optimizaiton problem, the automation can predict the vehicle's states and correct the driver's steering and braking actions to avoid the vehicle out of control while avoiding unnecessary interventions in a wide range of scenarios. The simulation results show that the proposed approach can effectively assist the driver to ensure safety and mitigate human-machine conflicts.

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